

Typcasting macros

If you look carefully at the code, you'll find some examples of typecasting of pointers to apparently different data types. For example, `GtkRadioButton` objects are sometimes typecast to `GtkToggleButton`s, and sometimes to `GtkButtons`, using macros. In `set_page_editable()`, all widgets are typecast to `GtkWidget` when passed as a parameter to `gtk_widget_set_can_focus()`. Such typecasts reveal the object-oriented nature of GTK+: `GtkRadioButton` derives from `GtkCheckButton`, which derives from `GtkToggleButton`; all the widgets used are derived from `GtkWidget`. This means that all widgets inherit properties and methods from their parent object hierarchy.

Macros like `GTK_WIDGET()` are used for typecasting any type of widget to a `GtkWidget`. Others like `GTK_TOGGLE_BUTTON()`, `GTK_BUTTON()`, etc., typecast widget pointers to the corresponding type after checking for the parameter's actual type. These macros make use of the advanced functionality of GLib and GCC, and can issue warnings in case of type mismatch. Using these macros is recommended over simple typecasts to the corresponding types. You can find them in the corresponding header files of widgets (for example, `GTK_BUTTON` is defined in `gtk/button.h`).

This is an example of the careful object-oriented design of GTK+, which makes it superior to many similar tool-kits. This also means that if a method you are looking for is not found in an object, you need to search all objects in the hierarchy before implementing it on your own.

is programmed not to show its tabs, by calling `gtk_notebook_set_show_tabs()` with a `FALSE` parameter. The singly-linked list `priv->gender` is populated with group information received, using `gtk_radio_button_get_group()`. Using this group information, radio buttons are more easily accessed in a list.

As an example of dynamic widget creation and attachment, create a combo-box using the function `gtk_combo_box_new_text()`. This function creates an empty combo box without any list items. First insert the text 'Level-2', using the insert function. Then, as an example of appending and pre-pending a list, you can modify the combo-box pull-down menu with two more entries. Although we have not used it, there is a function to remove items from the list as well. Please note the typecasting macros used in passing the parameter to these functions. Once the combo-box is created and modified, it is attached to the table container object using `gtk_table_attach_defaults()`.

As a final step, use `gtk_widget_show()` on the top level window, and call `gtk_main()` to enter the loop. This loop exits only when you quit the application. Then, dereference the `GtkBuilder` object, `g_free()` the

`g_malloc`-ed variable, and return 0.

Handlers

Just like its Perl implementation, you define the handler functions. Note that the handler functions have the same names as the Perl functions, since they are independent of the language of implementation—because they are defined in the `GTKBuilder` file created by Glade.

Quit and delete handlers

When the top level window is closed, the delete event handler (`on_window1_delete_event()`) is called. If this function returns `FALSE` (as we do), the delete event will propagate, and eventually calls `on_window1_destroy()`. This function calls `gtk_main_quit()` and makes the `gtk_main()` loop quit. The handler `on_quit_mni_activate()`, which is invoked when the `File->Quit` menu item is clicked, does the same. Once `gtk_main()` quits, the remaining part of the `main()` function will execute, performing a clean-up.

OK and Cancel buttons: login screen

An important thing to note about the handler functions is that all of them are

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